

Customizable beam splitting in planar adiabatic acoustic couplers composed of cylindrical scatterers

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ABSTRACT:

In this work, five-waveguide (five-WG) acoustic couplers with planar configurations are designed based on quantum-like adiabatic transfer, through which the incident waves can efficiently transfer from the middle WG to the other two WGs with a customized intensity ratio. The five WGs are connected by space-varying cylindrical scatterers, and the coupling between two adjacent WGs is determined by two Gaussian pulses with a certain delay. Since the evolution process of acoustic waves can adiabatically follow the dark state, the coupler could have a broadband and stable performance. Moreover, it is easy to change the ratio of the beam splitting by utilizing different peak values of the coupling between the middle three WGs. The agreements between analytical, numerical, and experimental results confirm the feasibility of the design, providing an effective solution for high-performance acoustic beam splitters with customizable output intensities. © 2024 Acoustical Society of America. <https://doi.org/10.1121/10.0028169>

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I. INTRODUCTION

Acoustic beam splitting is of great significance in many applications, including acoustic communication, sensing, and detection.^{1–4} To split the incident wave into multiple parts with desirable intensities, numerous available approaches have been reported in the past several years.^{5–9} For instance, the acoustic waves can propagate along the line defect in phononic crystals owing to the bandgap induced by periodic structures.^{10–12} Consequently, a beam splitting is easy to be realized by designing multiple appropriate channels with specific media. However, the performance of the device is sensitive to the frequency variation, which is attributed to the fact that the incident wavelength needs to be matched with the channel size. As another effective manner to achieve beam splitting, an acoustic metasurface composed of resonance units can manipulate the transmission fields following the generalized Snell's law.^{13,14} By taking advantage of the phase engineering, the acoustic wavefront incident from one side of the metasurface is able to be reconstructed to a particular beam with desired intensities and directions.^{15–23} Nevertheless, to meet the requirement of resonance response, the device configuration is always complicated and the working band is relatively narrow. Therefore, it is necessary to explore a new approach to achieve a customizable beam splitter with excellent performance.

As a kind of classical wave, the acoustic wave has shown great potential to realize quantumlike behaviors recently. Many fascinating quantum effects, such as non-Abelian braiding,^{24,25} the spin Hall effect^{26–28} and nonadiabatic holonomic quantum transformation²⁹ have been visually simulated and observed in acoustic systems. Among numerous quantum technologies, adiabatic passage has been widely studied currently due to its unique ability to obtain efficient and selective transfer between quantum states, which can be adopted to achieve one-way acoustic energy localization by performing an analogy between quantum and acoustic systems.^{30–33} Owing to the resonance-independent response, the energy propagation following the adiabatic passage is robust and stable. Hence, it is promising to design planar acoustic beam splitters based on quantumlike adiabatic transfer.

In this work, we intend to realize customizable beam splitting in five-waveguide (five-WG) acoustic couplers. Due to the agreement in forms between the Schrödinger equation and coupled mode equation, the acoustic waves propagating along the coupler could mimic the quantumlike adiabatic transfer. The coupling between adjacent WGs is modulated by space-varying cylindrical scatterers. Thanks to the pluggable feature of the scatterers, the coupling strength can be adjusted by directly changing the arrangement order of the scatterers on the base. Both numerical and experimental results show that the waves input from the left port of the middle WG can be output from the right ports of two-sided WGs with arbitrary intensities. The ratio of the

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splitting beam is determined by the interaction strength between adjacent WGs, and the transfer process of acoustic energy is robust against the incident wavelength in a certain range. Given the excellent performance of the adiabatic acoustic coupler, this work provides a new scheme for the design of advanced passive acoustic devices.

II. METHODS

A. Adiabatic propagation of acoustic waves

The schematic of the WG coupled acoustic system is shown in Fig. 1(a), the five straight WGs are connected by cylindrical scatterers with period $T = 7$ mm. The length, width, and height of the WG are $L = 900$ mm, $W_1 = 10$ mm, and $H = 6$ mm, respectively. The slit width is $W_2 = 7$ mm. As shown in Fig. 1(b), the expected energy transfer behavior is that the acoustic waves incident from the left port of WG 3 could propagate along the coupler adiabatically, and then a customizable beam splitting is realized from WGs 1 and 5 efficiently. To understand the quantumlike adiabatic acoustic transfer process, the five-WG acoustic coupler can be treated as a five-level quantum system.³³ Similar to the form of a Hamiltonian in the time dimension, the mapped coupling matrix of the acoustic system in the space dimension can be expressed as

$$M(x) = \begin{pmatrix} 0 & C_1(x) & 0 & 0 & 0 \\ C_1(x) & 0 & C_2(x) & 0 & 0 \\ 0 & C_2(x) & 0 & C_3(x) & 0 \\ 0 & 0 & C_3(x) & 0 & C_4(x) \\ 0 & 0 & 0 & C_4(x) & 0 \end{pmatrix}, \quad (1)$$

where $C_i(x)$ ($i = 1, 2, 3, 4$) represents the space-varying coupling strength between WGs m and $m+1$ ($m = 1, 2, 3$). By setting $C_1(x) = C_4(x)$ and $C_2(x) = C_3(x)$, the quantities $C_1(x)$ and $C_2(x)$ are similar to the Rabi frequencies for the pump and Stokes pulses in the stimulated Raman adiabatic passage in quantum physics. The system has an eigenvalue of $E(x) = 0$, and its corresponding eigenstate is the so-called dark state,

$$B_1(x) = \frac{1}{\sqrt{2}} \sin \theta |1\rangle - \cos \theta |3\rangle + \frac{1}{\sqrt{2}} \sin \theta |5\rangle, \quad (2)$$

where the mixing angle is $\theta(x) = 2\sqrt{C_2(x)}/C_1(x)$, which represents the relationship between two space-varying coupling strengths. Following the dark state expressed in Eq. (2), the mixing angle needs to vary from 0 to $\pi/4$ to obtain the result of beam splitting from WG 3 to WGs 1 and 5. The adiabatic coupling matrix can be deduced as

$$M_a(x) = \begin{pmatrix} 0 & 0 & -id\theta/\sqrt{2}dx & 0 & 0 \\ 0 & C_1(x) & 0 & 0 & 0 \\ id\theta/\sqrt{2}dx & 0 & \sqrt{C_1(x)^2 + 2C_2(x)^2} & 0 & -id\theta/\sqrt{2}dx \\ 0 & 0 & 0 & -C_1(x) & 0 \\ 0 & 0 & -id\theta/\sqrt{2}dx & 0 & -\sqrt{C_1(x)^2 + 2C_2(x)^2} \end{pmatrix}. \quad (3)$$

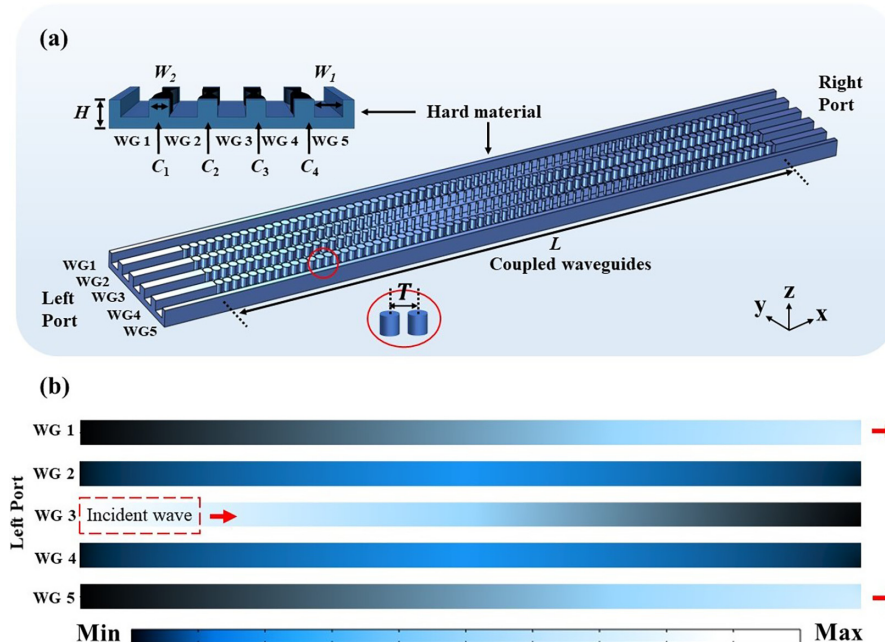


FIG. 1. (Color online) WG coupled acoustic system. (a) Schematic of the five-WG coupler with space-varying cylindrical scatterers. (b) Schematic of the energy transfer process.

To ensure the evolution process of acoustic waves following the dark state, the transitions between adiabatic states need to be eliminated. Consequently, the difference between the diagonal elements in Eq. (3) should be much larger than the off diagonal elements, and thus the adiabatic condition is obtained as

$$|d\theta/dx| \ll |C_1(x)|, |d\theta/dx| \ll \left| \sqrt{C_1(x)^2 + 2C_2(x)^2} \right|. \quad (4)$$

According to Eq. (4), it can be deduced that the trends of the coupling between adjacent WGs require changing slowly.

As a remarkable feature of the dark state [Eq. (2)], the states $|1\rangle$ and $|5\rangle$ have the same amplitude. Hence, the incident acoustic energy will transfer from WG 3 to WGs 1 and 5 once the mixing angle changes from 0 to $\pi/4$ smoothly. Note that the peak values of the space-varying coupling actions $C_2(x)$ and $C_3(x)$ are identical in this case, resulting in an equal-weighted beam splitting. By altering the peak values of $C_2(x)$ and $C_3(x)$, the beam splitting ratio can be customized flexibly.

Taking 2:1 beam splitting as an example, the peak value of $C_2(x)$ is set to be $\sqrt{2}$ times higher than that of $C_3(x)$ to achieve the target transfer process. In this case, the dark state of the system is expressed as

$$B_2(x) = \frac{\sqrt{2}}{\sqrt{3}} \sin \theta_2 |1\rangle - \cos \theta_2 |3\rangle + \frac{1}{\sqrt{3}} \sin \theta_2 |5\rangle, \quad (5)$$

where $\theta_2(x) = \sqrt{6}C_2(x)/2C_1(x)$. When $\theta_2(x)$ changes from 0 to $\pi/4$, the incident acoustic energy will adiabatically transfer from WG 3 to WGs 1 and 5 with an intensity ratio of 2:1. The splitter with other output ratios can be customized as well by selecting appropriate peak values of the coupling strength.

B. Coupling modulation between adjacent WGs

To modulate the coupling between adjacent WGs, as shown in Fig. 2(a), a two-WG acoustic coupler composed of periodic cylindrical scatterers is designed. All cylindrical scatterers have the same size. The coupling strength between WGs can be controlled by using different values of the scatterer radius. To obtain the acoustic intensity field of the coupler, a numerical simulation is performed by the software COMSOL MULTIPHYSICS. In the numerical simulations, the material of the background medium is selected as air (the density is $\rho_{\text{air}} = 1.21 \text{ kg/m}^3$ and the acoustic velocity is $c_{\text{air}} = 343 \text{ m/s}$). Owing to the huge impedance difference between rigid bodies and air background, the rigid materials can be set as hard boundary conditions in the numerical simulations. In practical application, the rigid materials can be selected as epoxy resin ($\rho_m = 1100 \text{ kg/m}^3$ and $c_m = 2540 \text{ m/s}$). All the input and output ports are set as planar wave radiation conditions, and the acoustic pressure value of the incident plane wave is set as 1 Pa. As illustrated in Fig. 2(b), the acoustic intensity oscillates back and forth inside the coupler for WG 1 incidence at

10 kHz. The distance between two adjacent peaks of the intensity field along each WG is defined as the coupling length d , which is closely related to the cylinder radius r . According to the simulated results, it is seen that the coupling lengths are $d = 300.2 \text{ mm}$, $d = 198.75 \text{ mm}$, $d = 163.57 \text{ mm}$, and $d = 143.28 \text{ mm}$ for cases of $r = 3 \text{ mm}$, $r = 2.5 \text{ mm}$, $r = 2 \text{ mm}$, and $r = 1.5 \text{ mm}$. Since the coupling strength C is inversely correlated with the coupling length ($C = \pi/d$), it can be concluded that C can be relatively high under a smaller cylinder radius r , which is attributed to the fact that the widths of the air slit between two WGs increase with the decrease in r under a fixed period T . By scanning the radius of the scatterer from 1 mm to 3.4 mm (interval of 0.1 mm), several discrete values of C are obtained [shown in Fig. 2(b)], and then the relationship between the coupling strength and radius can be fitted by a quartic function of $r = 2.34(C/C_{\text{max}}) - 11.63(C/C_{\text{max}})^2 + 14.61(C/C_{\text{max}})^3 - 7.55(C/C_{\text{max}})^4 + 3.28$, where $C_{\text{max}} = 23.23 \text{ m}^{-1}$ refers to the coupling strength when $r = 1 \text{ mm}$. Therefore, the coupling between two adjacent WGs is capable of modulating by utilizing different structural parameters of the cylindrical scatterer. Since the periodic length of the cylindrical scatterer is $T = 7 \text{ mm}$ and the minimum machinable radius is $r = 0.1 \text{ mm}$, the available range of the radius can be changed from 0.1 mm to 3.4 mm. Following the relationship between the coupling strength and r , the range of C_{max} can be calculated as 4.24 m^{-1} – 25.72 m^{-1} .

C. Simulation and analytical analysis of couplers

To fulfill the requirement of mixing angle [$\theta(x)$ changes from 0 to $\pi/4$ smoothly], as shown in aquamarine and lavender shaded areas of Fig. 3(a), two space-dependent normalized Gaussian pulses $C_1(x) = e^{-(x-x_0)^2/\delta^2}$ and $C_2(x) = e^{-((x-x_0)+s)^2/\delta^2}$ with $x_0 = 430 \text{ mm}$, $s = 55 \text{ mm}$, and $\delta = 134 \text{ mm}$, are adopted to determine the coupling between adjacent WGs, which is inspired by the forms of Rabi frequencies for pump (Ω_p) and Stokes pulses (Ω_s) in the stimulated Raman adiabatic passage in quantum physics. For an equal-weighted beam splitting, we set $C_1(x) = C_4(x)$ and $C_2(x) = C_3(x)$, and the corresponding space-varying radii are given by aquamarine and lavender dotted lines. To verify the feasibility of our design, the acoustic intensity field of the five-WG coupler illustrated in Fig. 3(b) is numerically simulated at 10 kHz. As shown in Fig. 3(c), the simulated results (point lines) of the normalized acoustic intensity distribution along each WG agree well with the analytical results (solid lines) calculated by the Schrödinger-like equation

$$i \frac{d}{dx} \psi(x) = M(x) \psi(x), \quad (6)$$

where $\psi(x) = [\psi_1(x), \psi_2(x), \psi_3(x), \psi_4(x), \psi_5(x)]^T$ with $\psi_k(x)$ being the acoustic amplitude along WG k ($k = 1, 2, 3, 4, 5$). From the simulated acoustic intensity fields illustrated in Fig. 3(d), it is observed that the acoustic waves can transfer from WG 3 to WGs 1 and 5 with identical intensities, exhibiting an

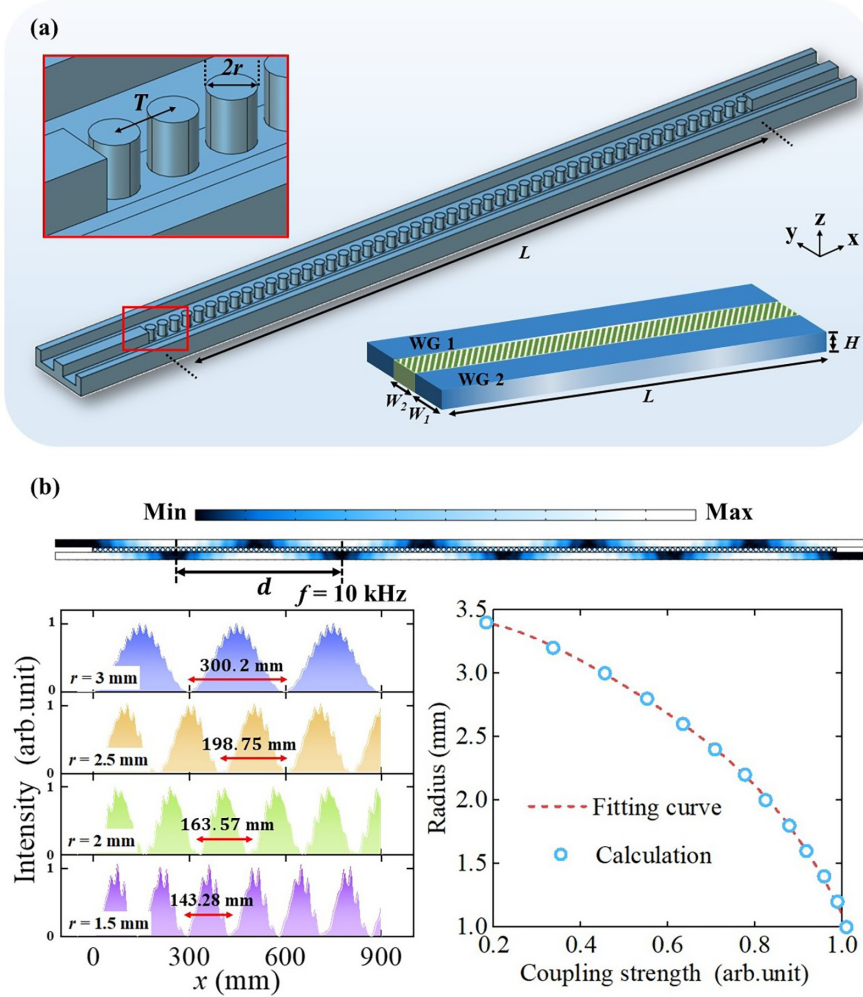


FIG. 2. (Color online) Coupling between adjacent WGs. (a) Schematic of the two-WG acoustic coupler. (b) The discrete and fitting process of the coupling strength. arb. unit, arbitrary unit.

excellent beam splitting behavior in the planar acoustic WG coupler. For other ratios of beam splitting, it is only necessary to customize the peak values of $C_2(x)$ and $C_3(x)$. As shown in Fig. 3(e), we set $C_1(x) = C_4(x)$ and $C_2(x) = \sqrt{2}C_3(x)$ to design the beam splitter shown in Fig. 3(f). The simulated results of normalized acoustic intensity distribution along each WG illustrated in Fig. 3(g) are consistent with the analytical results as well, and the simulated acoustic intensity fields illustrated in Fig. 3(h) further verify the generation of beam splitting with a ratio of 2:1.

Actually, the adiabatic acoustic coupler could have a broadband feature since it follows the mechanism of coupling interaction rather than resonance process. As shown in Fig. 4(a), an equal-weighted beam splitting still can be generated at three operating frequencies with large spans (7.2, 8.5, and 10.5 kHz). The frequency spectra of the output intensities from WGs 1 and 5 are plotted in Fig. 4(b): it can be found that the equal-weighted beam splitting can be generated in a ultrabroadband of 6–16 kHz, and the energy transfer efficiency is higher than 0.8 in a wide range from 7.2 kHz to 14.2 kHz. Meanwhile, the efficiency can be further improved (>0.9) in a broadband of 8.5–12.5 kHz (marked by the light blue area), and the peak value [$I_1 = I_5 = 0.491$, where I_k represents the normalized acoustic

intensity of WG k ($k = 1, 2, 3, 4, 5$)] appears at 10.5 kHz, which is close to the central discrete frequency (10 kHz).

Similarly, by customizing the coupling with $C_2(x) = \sqrt{2}C_3(x)$, as illustrated in Fig. 4(c), a beam splitting with ratio of 2:1 can be produced at multiple operating frequencies with large spans such as 7.4 kHz ($\eta = 0.85$), 9.0 kHz ($\eta = 0.92$), and 10.2 kHz ($\eta = 0.99$) as well. According to the frequency spectra of the output intensities from WGs 1 and 5 shown in Fig. 4(d), it can be observed that the energy transfer efficiency is higher than 0.8 in a broadband of 7.4–13.6 kHz (marked by the green area). Within this band, the normalized intensities of output waves from WGs 1 and 5 are around 0.6 and 0.3 and the peak value appears at 10.2 kHz ($I_1 = 0.637$, $I_5 = 0.352$). The slight frequency difference of two couplers with different beam splitting behaviors is attributed to the imprecise relationship between the radius and the coupling strength, which is hardly to be completely avoided due to the existence of strong intensity oscillation in simulated acoustic fields.

Although the proposed acoustic system provides an effective solution for the design of acoustic coupler for beam splitting, there is a limitation in the current scheme. The precision of a discrete cylindrical scatterer should be

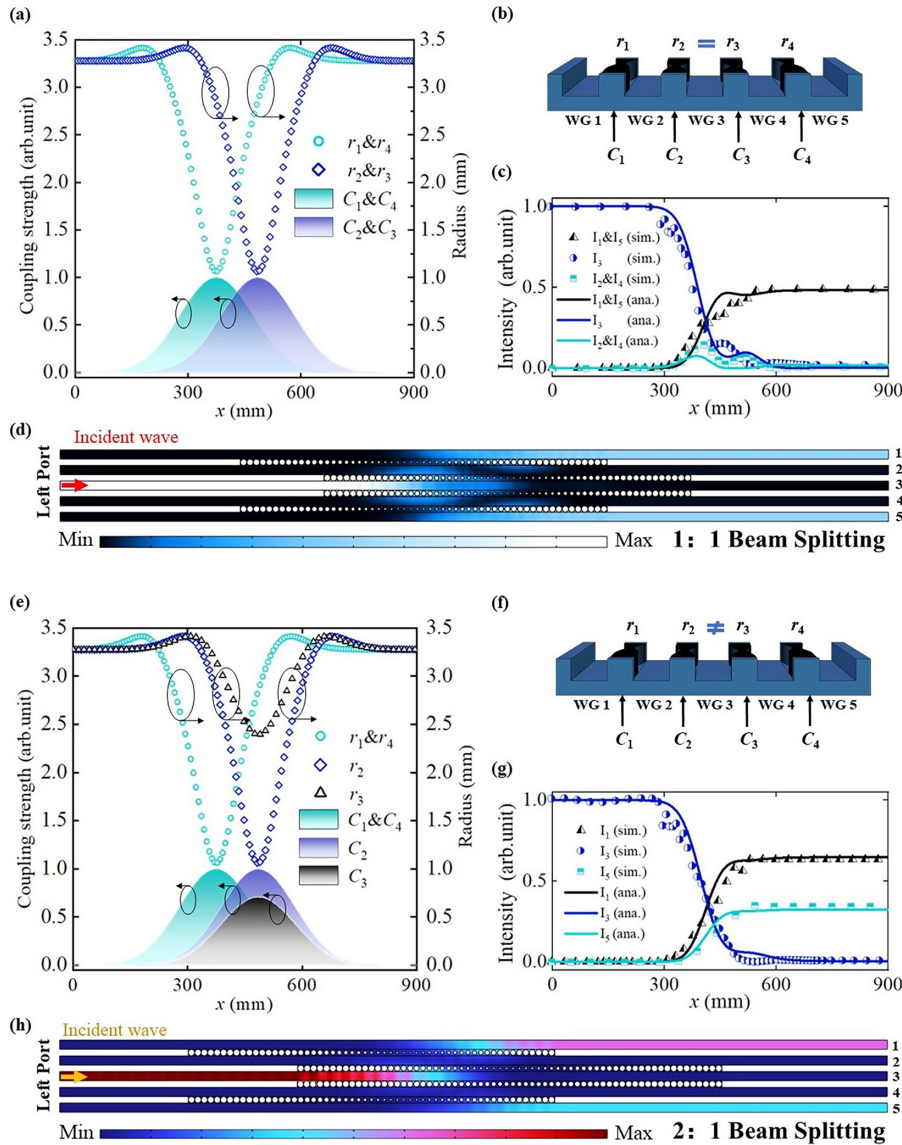


FIG. 3. (Color online) The space-varying coupling strengths and corresponding radii along the propagation direction for (a) equal-weighted and (e) 2:1 beam splitting. (b) and (f) Side view of the five-WG acoustic coupler. (c) and (g) Simulated and analytical normalized acoustic intensity distributions along the propagation direction of the five WGs for (c) equal-weighted and (g) 2:1 beam splitting. (d) and (h) Simulated acoustic intensity fields at 10 kHz for (d) equal-weighted and (h) 2:1 beam splitting.

considered while mapping the driving pulses into parameter space. The period length and maximum radius of the cylindrical scatterer need to be as short as possible to guarantee enough discrete points in a fixed device length. On the other hand, in actual processing, the resin three-dimensional (3D) printing process imposes limitations on the radius of the cylindrical scatterer. To adhere to these constraints, it is advisable for the radius to satisfy the relationship $r \geq 1$ mm as closely as possible. Consequently, the acoustic coupler designed in our work is a compromise consideration.

The acoustic WG coupler reported by Shen *et al.* can realize a desirable energy transfer from one WG to another.³⁴ Compared with their work, the five-WG coupler proposed in our work has more sophisticated features. Not only can it show an excellent beam splitting performance with customizable ratios, but it also has richer energy transfer behavior results. For example, the power splitting is able to be realized from WG 1 to WGs 3 and 5 as well by altering the coupling actions between adjacent WGs.

III. EXPERIMENTAL VERIFICATION

Next, we intend to experimentally verify the performance of the coupler. To manifest the flexible feature of the design, the intensity ratio of two splitting beams is customized as 3:2, which requires that the coupling $C_2(x) = \sqrt{6}C_3(x)/2$. The simulated intensity field at 10 kHz is shown in Fig. 5(a); acoustic waves incident from the left port of WG 3 can be transferred into the right ports of WGs 1 and 5 successfully. The output intensities of WGs 1 and 5 are calculated as $I_1 = 0.591$ and $I_2 = 0.405$, respectively, which are nearly coincide with the analytical results shown in Fig. 5(b). Owing to the planar configuration of the beam splitter, it is easy to be fabricated by a 3D printing technique. Photosensitive resin, which has a significant impedance mismatch with the air background, is chosen as the rigid material. For sample preparation, it is essential to fabricate a base of the coupler composed of five straight WG channels and several reserved cylindrical sockets. As illustrated in Fig. 5(d), the base can be reused for the realization of various beam splitters, and the target intensity ratio of the output

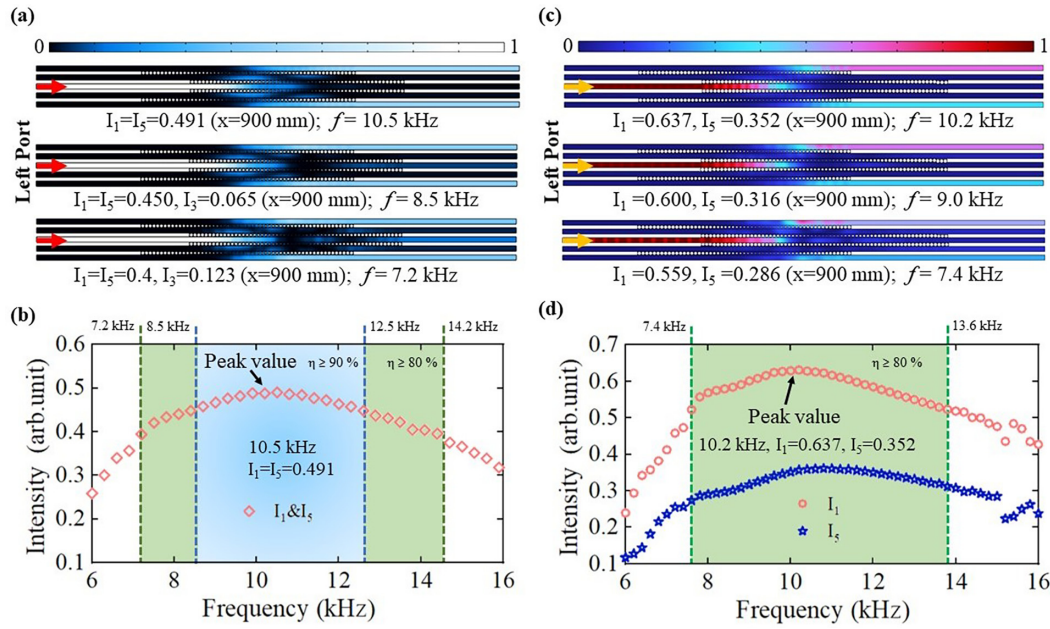


FIG. 4. (Color online) Acoustic intensity fields of the coupler for (a) equal-weighted and (c) 2:1 beam splitting. (b) and (d) The working bands of the coupler for (b) equal-weighted and (d) 2:1 beam splitting.

waves is customized by the space-varying cylindrical scatterers. The structural parameters are the same as the ones utilized in numerical simulations. For the convenience of measurement, a rigid plate with periodic holes (the sizes are matched with microphones) is fabricated as well to detect the intensity responses at certain positions along five WGs.

As shown in Fig. 5(e), an experimental platform is built to measure the acoustic intensity distributions of the coupler. The incident waves are generated by a loudspeaker shown in Fig. 5(f), and the frequency and amplitude can be controlled by the signal generator and power amplifier. The intensity responses are detected by inserting the microphones into the unbolted holes on the cover plate. Note that the unscanned holes need to be bolted by caps to avoid the leakage of transmission waves. The measured results are recorded in a data acquisition device and processed by a signal analysis software, as shown in Fig. 5(g), it can be found that the normalized acoustic intensities are highly localized in WG 3 at the input port, while those are different at the output ports of WGs 1 and 5. Upon setting the incident frequency to 10 kHz, an experimental outcome closely approaching ideal beam splitting in a 3:2 ratio is observed. This outcome aligns closely with both analytical predictions and simulations. Additionally, although the output intensities are slightly changed at other frequencies (such as 8.2 and 13 kHz), the ratios of beam splitting can remain constant at around 3:2, verifying the feasibility of the acoustic planar beam splitter designed by adiabatic passage.

Note that the beam splitter with arbitrary ratios can serve as an acoustic router and may have potential applications in acoustic communications. In particular, the device could provide a one-way communication capability. Following the dark state expressed in Eq. (5), it can be deduced that the WGs 2

and 4 are not involved in the evolution process. Consequently, as illustrated in Fig. 3(d), for the left side of WG 3 incidence, the acoustic intensities are highly localized along WGs 1, 3, and 5, while those along WGs 2 and 4 are relatively weak. For the right side of WG 3 incidence, however, the energy transfer behavior is quite different from that for the left side of WG 1 incidence. The propagation path of the acoustic energy will follow the complex superposition states, and the high acoustic intensities will distribute along WGs 2 and 4. Taking the beam splitting of 3:2 as an example, as shown in Fig. 5(c), one-way acoustic beam splitting is realized by introducing a strong dissipation along WGs 2 and 4. At this time, the acoustic wave incident from the left port of WG 3 is not affected seriously, while that incident from the right port of WG 3 is dissipated along the coupler. Hence, the acoustic coupler proposed in this work can be applied to realize a one-way communication.

IV. CONCLUSION

In conclusion, an approach to realize customizable acoustic beam splitting is proposed based on quantumlike adiabatic transfer. The acoustic waves propagating along the five-WG acoustic coupler can mimic the evolution process of quantum states. Two Gaussian pulses with a certain delay are adopted to modulate the space-varying coupling between adjacent WGs by taking advantage of the cylindrical scatterers with different diameters. The arbitrary ratio of the beam splitting can be customized easily by altering the peak values of the coupling strengths between WGs 2 and 3 and WGs 3 and 4, which has been verified by analytical calculations, numerical simulations, and experimental measurements. Meanwhile, the adiabatic acoustic coupler could remain at high efficiency in a wide frequency range from

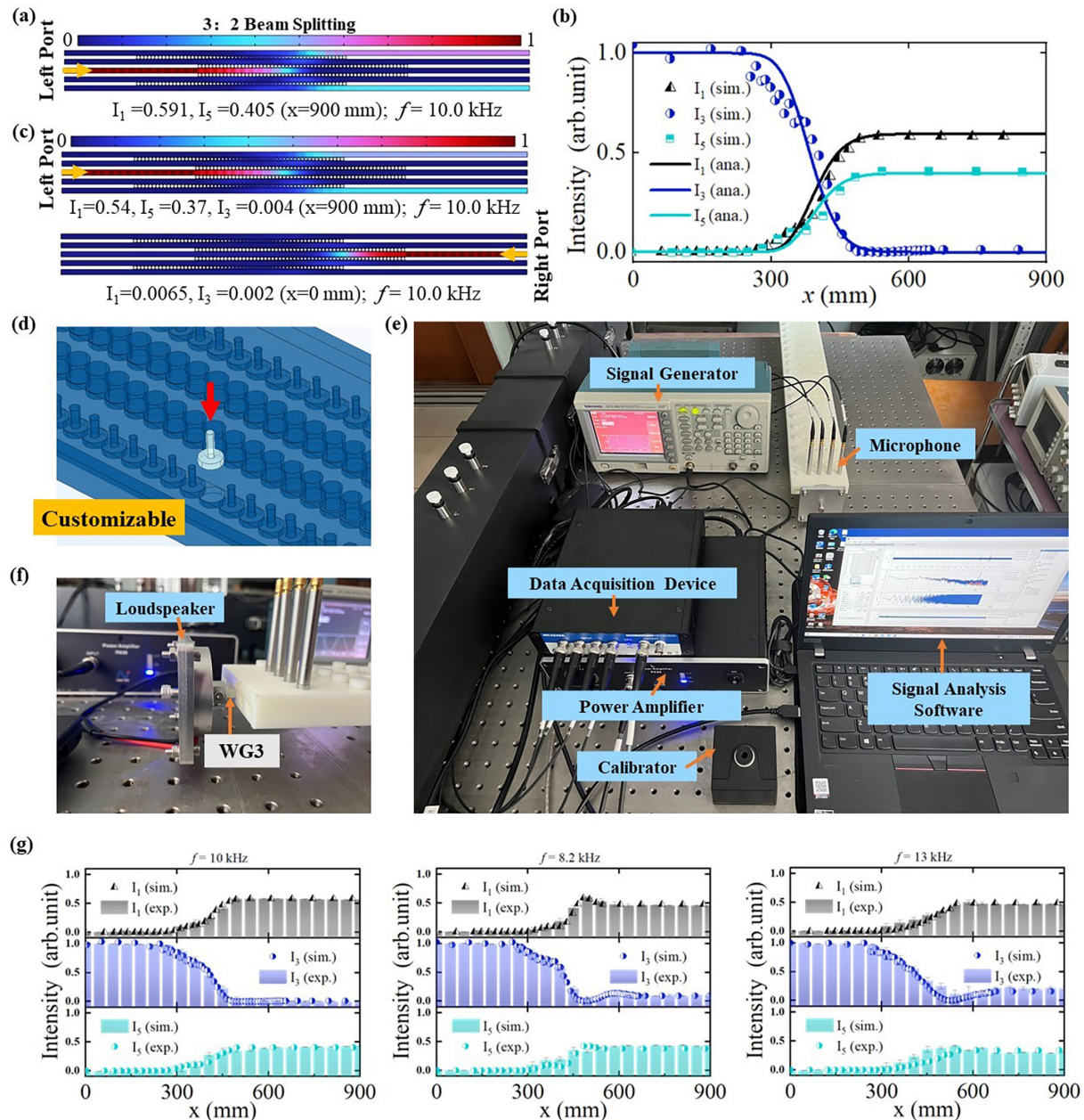


FIG. 5. (Color online) Design and experimental verification of the coupler for 3:2 beam splitting. (a) Simulated acoustic intensity field of the coupler for 3:2 beam splitting. The working frequency is 10 kHz, and the acoustic waves are incident from the left port of WG 3. (b) Simulated (sim.) and analytical (ana.) normalized acoustic intensity distributions along the propagation direction of the WGs 1, 3, and 5 for 3:2 beam splitting. (c) Simulated acoustic intensity field of the lossy coupler for one-way beam splitting. (d) Schematic of the customizable cylindrical scatterers. (e) Experimental platform for acoustic coupler measurement. (f) The detail of the input port and loudspeaker adopted in the measurement. (g) Measured results of the normalized acoustic intensity along WGs 1, 3, and 5 for WG 3 incidence at 8.2, 10, and 13 kHz.

8.5 kHz to 12.5 kHz, exhibiting an excellent and stable working performance. Given the advantages of customizable feature, broadband response, simple fabrication as well as planar configuration, the beam splitter proposed in this work may have potential applications in acoustic detection, sensing and communication.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts of interest to declare. All co-authors have seen and agree with the contents of the manuscript and there is no financial interest to report. We certify that the submission is original work and is not under review at any other publication.

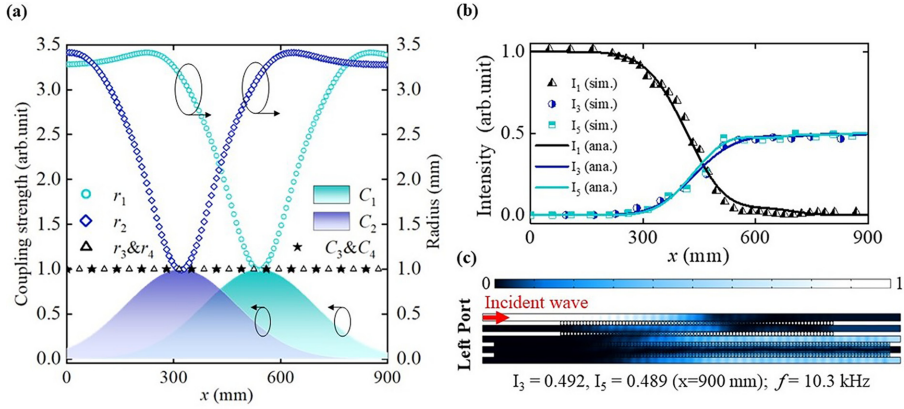


FIG. 6. (Color online) The space-varying coupling strengths and corresponding radii along the propagating direction. (b) The normalized acoustic intensities along the propagating direction. (c) Acoustic intensity field of the coupler at 10.3 kHz for the left port of WG 1 incidence.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: THE BEAM SPLITTING FROM WG 1 TO WGS 3 AND 5

The energy transfer behaviors can be customized by altering the coupling actions between five WGs. For example, if the acoustic energy is expected to be transferred from WG 1 to WGs 3 and 5, the coupling strength between WGs 3 and 4 can be set as the same value as that between WGs 4 and 5. In this case, the coupling matrix can be expressed as

$$M(x) = \begin{pmatrix} 0 & C_1(x) & 0 & 0 & 0 \\ C_1(x) & 0 & C_2(x) & 0 & 0 \\ 0 & C_2(x) & 0 & C_g(x) & 0 \\ 0 & 0 & C_g(x) & 0 & C_g(x) \\ 0 & 0 & 0 & C_g(x) & 0 \end{pmatrix}. \quad (A1)$$

The mixing angle is $\theta(x) = \arctan(C_2(x)/\sqrt{2}C_1(x))$. The dark state of the system can be deduced as

$$B_g(x) = \sin \theta |1\rangle - \frac{1}{\sqrt{2}} \cos \theta |3\rangle + \frac{1}{\sqrt{2}} \cos \theta |5\rangle. \quad (A2)$$

Therefore, beam splitting from WG 1 to WGs 3 and 5 can be obtained by making the mixing angle $\theta(x)$ changes from

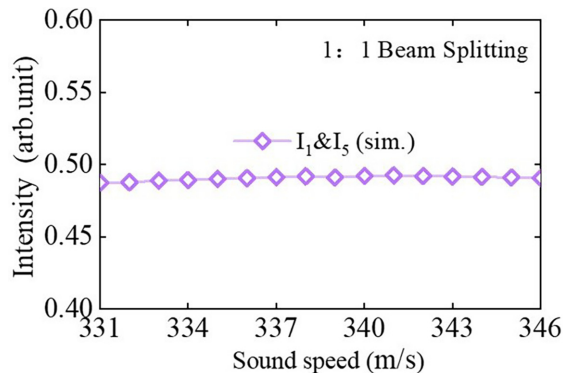


FIG. 7. (Color online) The relation between the output intensities and sound speeds.

$\pi/2$ to 0. Here, as illustrated in Fig. 6(a), the coupling strengths $C_1(x)$ and $C_2(x)$ are $C_1(x) = e^{-(x-x_0)^2/\delta^2}$ and $C_2(x) = e^{-((x-x_0)+s)^2/\delta^2}$ with $x_0 = 430$ mm, $s = 110$ mm, and $\delta = 220$ mm. The coupling strength $C_g(x) = 31 \text{ m}^{-1}$ is a constant. By mapping the coupling actions into the parameter space, as shown in Figs. 6(b) and 6(c), the acoustic waves incident from the left port of WG 1 are transferred to the right port of WGs 3 and 5 with a nearly equal proportion, which is consistent with the analytical results.

APPENDIX B: THE INFLUENCES OF SOUND SPEED AND ACTUAL MATERIAL ON THE PERFORMANCE OF BEAM SPLITTER

Owing to the robust feature of the adiabatic passage, the performance of beam splitting is stable even if the sound speed changes to other values (e.g., 331 m/s at 0°C –346 m/s at 25°C in air). For instance, consider the 1:1 beam splitter depicted in Fig. 3(d). As illustrated in Fig. 7, numerical simulations demonstrate that the beam splitter maintains stable and efficient energy transmission capabilities even when the sound speed varies from 331 m/s to 346 m/s.

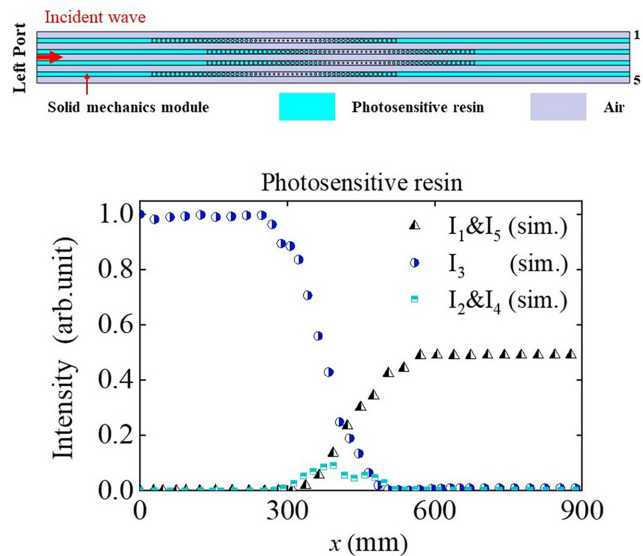


FIG. 8. (Color online) The schematic of the simulated acoustic model and the simulated acoustic intensities along the propagating direction of five WGs.

In addition, the beam splitter can still work when we replace the rigid boundary condition and consider the realistic photosensitive resin in numerical simulations. As shown in Fig. 8, the air background and the photosensitive resin material are connected by the “acoustic-structure boundary,” and the density, Young’s modulus, and Poisson’s ratio of the photosensitive resin are 1100 kg/m^3 , 2400 MPa , and 0.4 , respectively. Consider the 1:1 beam splitter depicted in Fig. 3(d). From the simulated acoustic intensities along the propagating direction of five WGs illustrated in Fig. 8, it can be observed that a desired equal-weighted beam splitting from WGs 1 and 5 is obtained for WG 3 incidence.

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